Type MS · Logical and arithmetic binary shifts

Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

Type MS · Logical and arithmetic binary shifts

Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · Logical and arithmetic binary shifts

Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · Logical and arithmetic binary shifts

Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · Logical and arithmetic binary shifts

Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Logical and arithmetic binary shifts

Answer 2 of 5 · Question 1 part e · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part e in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Logical and arithmetic binary shifts

Answer 2 of 5 · Question 1 part e · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part e in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Logical and arithmetic binary shifts

Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

Type MS · Logical and arithmetic binary shifts

Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · Logical and arithmetic binary shifts

Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · Logical and arithmetic binary shifts

Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · Logical and arithmetic binary shifts

Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Logical and arithmetic binary shifts

Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · Logical and arithmetic binary shifts

Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

Type MS · Logical and arithmetic binary shifts

Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · Logical and arithmetic binary shifts

Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Logical and arithmetic binary shifts

Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		2
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		1
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		2
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	1

Type MS · Logical and arithmetic binary shifts

Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(ii)</td><td>Award one mark for each of:

• 16x16 pixels (1)
• Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1)
• Dividing by 8 in the denominator to convert to bytes (1)
• Dividing 1000 by the number of bytes used for one image (1)

Example:

$1000 / \frac{16 \times 16 \times 2}{8}$ bytes

Allow follow-through</td><td>4 marks for any of:
1000 X 8 / 16 x 16 x 2
8000 / 256 x 2
1000 / 16 x 2 x 2
1000 / 16 x 4
8000 / 512</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(iii)</td><td>Award up to two marks for a linked explanation such as:

• high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1)
• requires less storage space (1) encoding 16W for line 1 (1)</td><td>Allow valid RLE examples from the image.

Explanation can be a combination of different marking points. E.g. MP1a and MP2b</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(c)</td><td>Award one mark for each correct cell:

<table><tr><th>Encrypted text</th><th>Plain text</th></tr><tr><td>  </td><td>RED</td></tr><tr><td></td><td>COT</td></tr></table></td><td>Ignore capitalisation
Must be correct spelling</td><td>2</td></tr>
<tr><td colspan="3">Total for question 3</td><td>14</td></tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Logical and arithmetic binary shifts

Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

Question	Answer	Additional Guidance	Mark